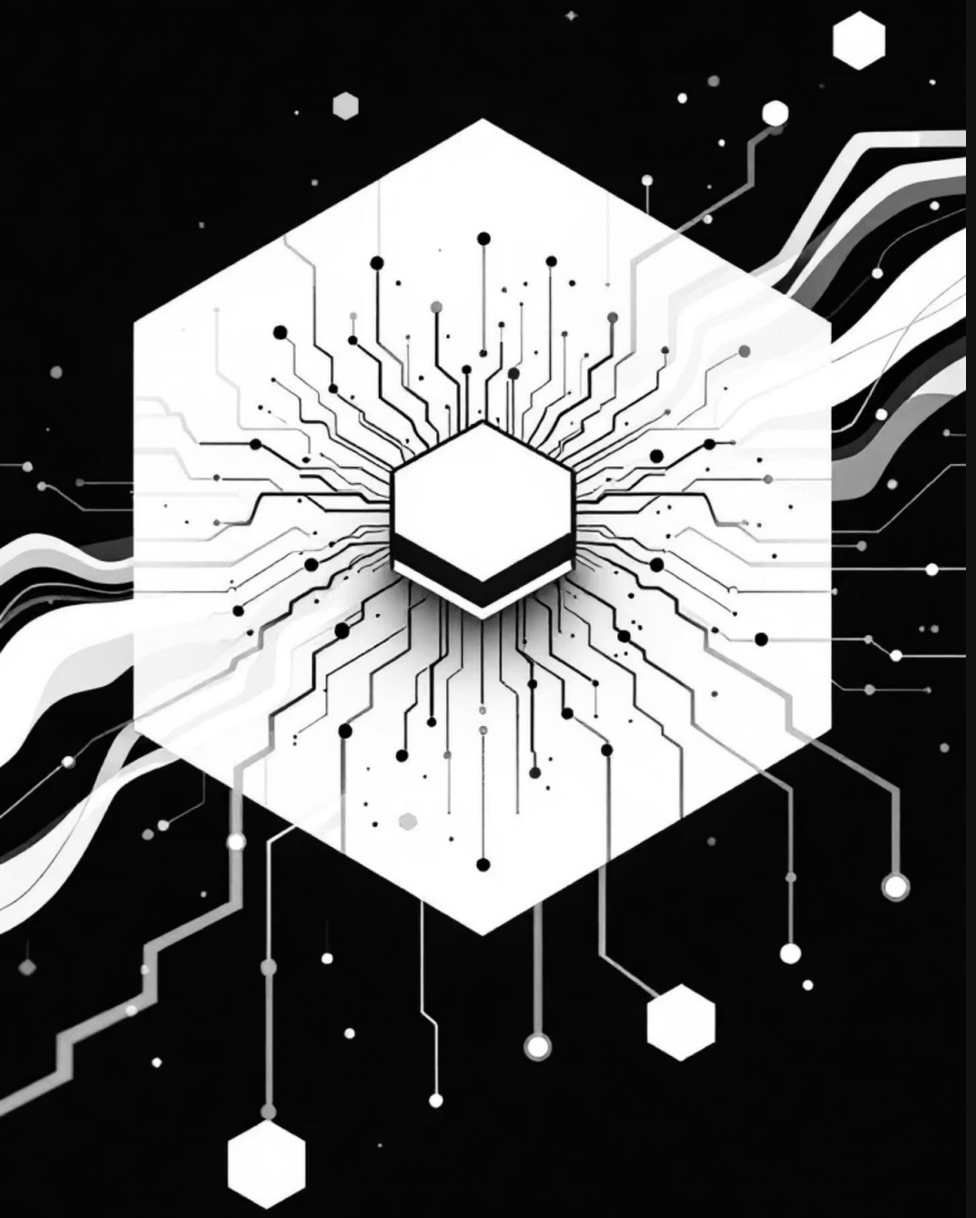


Types of Data and Data types

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What is Data?

Data is raw information, facts, or statistics that exists in various forms

- Numbers
- Text
- Images
- Audio
- Video

In its initial state, data is unprocessed and unstructured—it only becomes valuable "information" after we organize, analyze, and provide context.

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Understanding Data: The Foundation

Data is all around us, generated by people, systems, and the natural world and thus can be collected from various sources.

People-Generated Data

- **Surveys & Forms**: Customer feedback forms, online quizzes.
- **Social Media**: Likes, shares, posts, and comments.
- **Transactions**: Purchase histories, bank transfers, and e-commerce orders.

System-Generated Data

- **Sensors & IoT**: GPS location, temperature sensors, smart meter readings.
- **Web Servers**: Website traffic logs (clicks, page views, session duration).
- **Machines**: Industrial equipment performance metrics and error logs.

Traditional Sources

- **Public Records**: Government census data, economic reports, weather data.
- **Business Documents**: Invoices, reports, and sales receipts.



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Types of Data

Data can be classified in multiple ways depending on its structure, format, and the nature of information it represents.

By Structure & Format

- Structured Data
- Unstructured

By Nature & Analysis

- Quantitative
  - Discrete
  - Continuous
- Qualitative
  - Nominal
  - Ordinal





Classification by Structure & Format

Structured Data

Organized in a predefined model with rows and columns, making it highly searchable and easy to analyze.

- *Tabular format with fixed schema*
- *Relational databases and spreadsheets*
- *Examples: SQL tables, Excel files, CSV data*

Unstructured Data

No predefined structure or organization, requiring specialized tools and techniques for analysis and interpretation.

- *Various formats without fixed schema*
- *Multimedia and document files*
- *Examples: Text documents, images, audio files, video content*

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Examples of Structured

Structured Data

Organized in a predefined model with rows and columns, making it highly searchable and easy to analyze.

Table 2.1 Customer data sample.

custid	sex	is.employed	income	marital.stat	housing.type	num.vehicles	age	state.of.res
2068	F	NA	11300	Married	Homeowner free and clear	2	49	Michigan
2073	F	NA	0	Married	Rented	3	40	Florida
2848	M	True	4500	Never married	Rented	3	22	Georgia
5641	M	True	20000	Never married	Occupied with no rent	0	22	New Mexico
6369	F	True	12000	Never married	Rented	1	31	Florida

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Examples of Unstructured

Unstructured Data

Unstructured data lacks clear organization and labels, making it difficult for automated systems to process.

"A patient presented with symptoms including a fever of 102°F, a heart rate of 110 bpm, and a blood pressure reading of 150/95 mmHg. The individual also reported a persistent cough for five days."

The Problem:
This paragraph contains critical data points: 102, 110, 150/95, 5. However, they are embedded in raw text without labels. A computer program would struggle to accurately identify that:

- 102 is a fever
- 110 is a heart rate
- 150/95 is blood pressure
- 5 is the duration of a cough

Without this context, the data cannot be systematically analyzed or used by an algorithm, highlighting the need for processing to transform it into a structured format.

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Examples of Unstructured Cont…

- Videos
  - TikTok clips
- Images
  - James Webb Space Telescope imagery
  - Facial images
  - Handwriting samples
  - Road signs
- Audio
  - Alexa voice commands
  - Real-time translation
  - Music
- Biometrics
  - Fingerprints
  - Facial recognition data
- Haptics
  - Phone vibration alerts

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Semi-structured

Semi-structured Data

Has some organization but not in rigid tables

Graph data doesn't fit neatly into the simple "Structured/Unstructured" categories. It is a **specialized structural format** designed specifically to represent connections and relationships

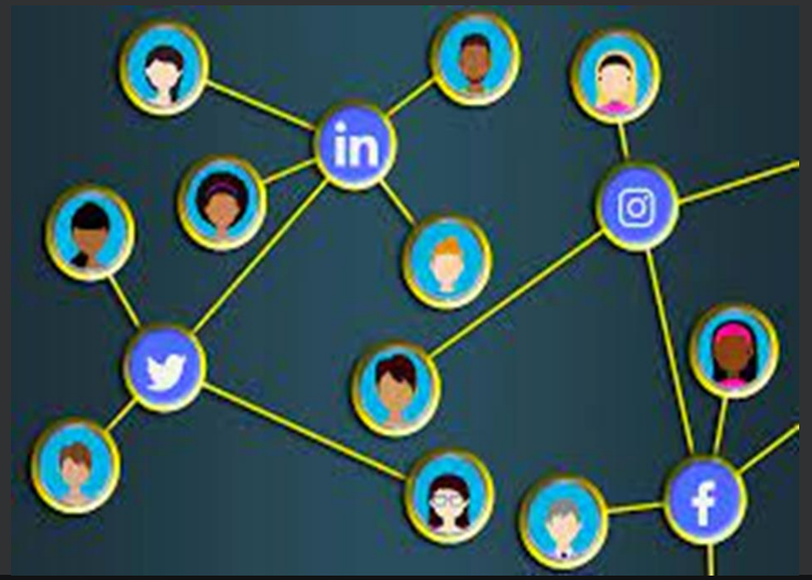
Characteristics of Graph Data:

Focus: Relationships and Networks

Components:

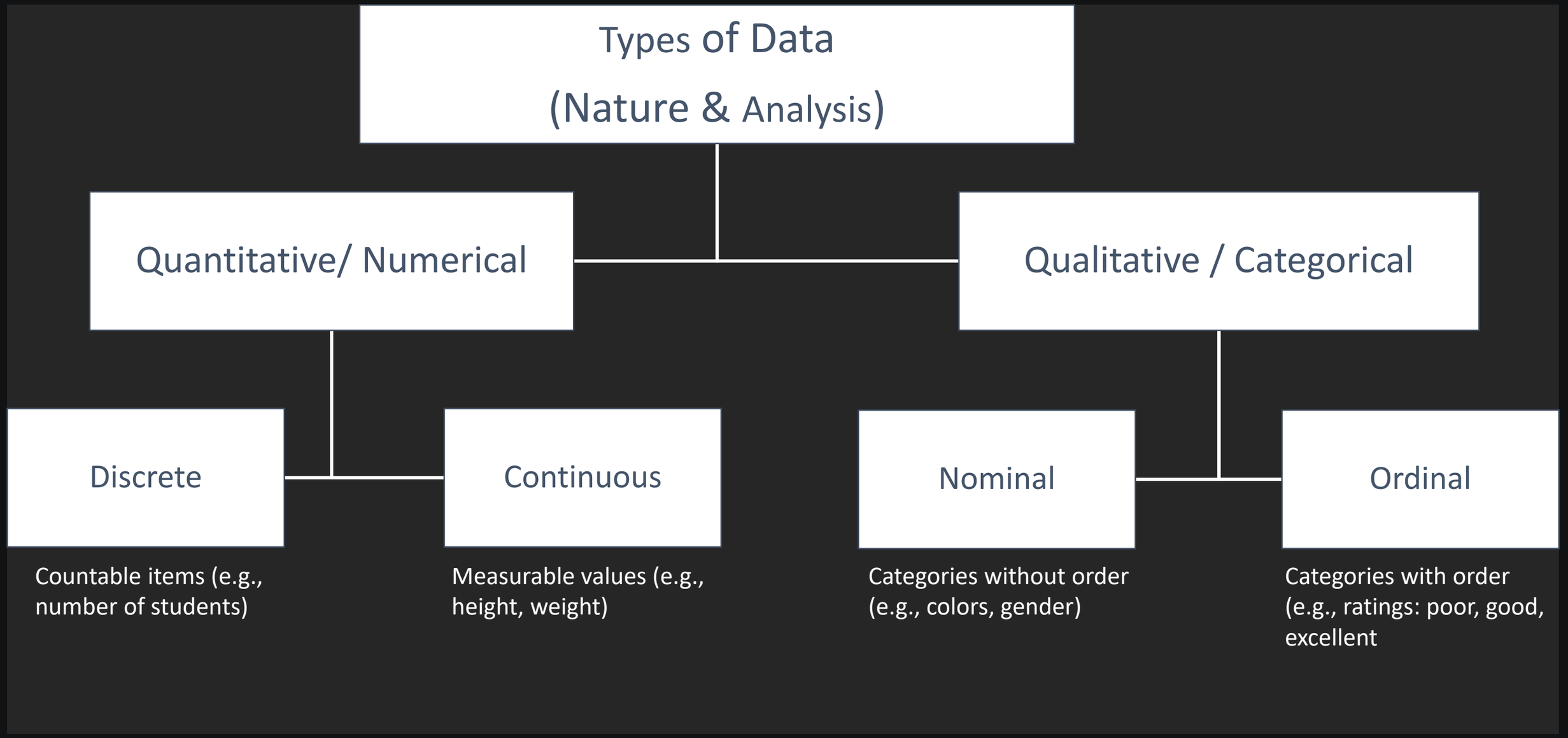
- Nodes (Vertices): Represent the entities or "things" (e.g., people, websites, cities).
- Edges (Connections): Represent the relationships or links between them (e.g., friendships, hyperlinks, roads).

Structure: Semi-structured. It has a defined model (nodes and edges) but is not tabular.



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Types of Data (Nature & Analysis)



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Numerical Data: Working with Quantities

Numerical data is expressed in numbers and can be measured mathematically. This type of data is fundamental to statistical analysis and data science, forming the backbone of quantitative research.

Discrete Data

Countable numbers that are distinct and separate—you can't have a fraction of one unit.

- *Number of students in a class (can't have 25.5 students)*
- *Cars in a parking lot*
- *Items sold in a store*
- *Number of employees*

Continuous Data

Measurable numbers that can take any value within a range and can be meaningfully divided.

- *Height (175.5 cm is meaningful)*
- *Weight and temperature*
- *Time duration*
- *Distance traveled*





Categorical Data: Organizing by Characteristics

Categorical data represents characteristics or groups, often non-numeric but essential for classification and segmentation in analysis.

Ordinal Data

Categories with a natural, meaningful order or rank that shows progression.

- *Customer satisfaction: Poor, Fair, Good, Excellent*
- *Education level: High School, Bachelor's, Master's, PhD*
- *Performance ratings: Below Average, Average, Above Average*

Nominal Data

Categories with no inherent order—simply different types or groups.

- *Colors: Red, Blue, Green*
- *Product types: Electronics, Clothing, Food*
- *Geographic regions: North, South, East, West*

Discrete Continuous Ordinal Nominal

Employee Performance & Details

Emp ID	Name	Age	Department	Height (cm)	Salary	Performance Rating
003	Alex S	26	Store	175.2	$55,250.75	Good
004	Gollin K	32	Marketing	182.9	$68,500.00	Excellent
005	Rabin R	31	Marketing	168.4	$65,175.50	Average
006	Jens M	26	Security	190.1	$48,885.25	Good
007	Maria T	29	Engineering	165.8	$82,340.60	Excellent

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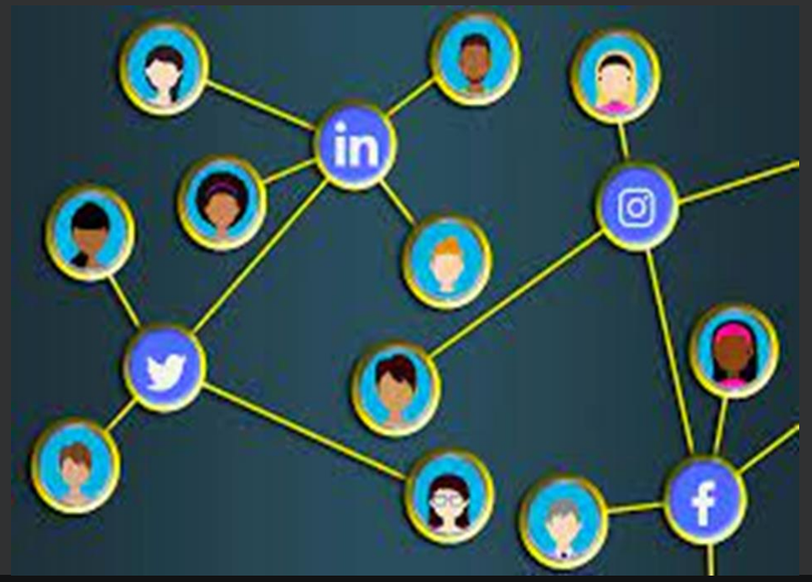
Semi-structured

Semi-structured Data

Has some organization but not in rigid tables

Graph data doesn't fit neatly into the simple "Quantitative/Qualitative" or "Structured/Unstructured" categories. It is a **specialized structural format** designed specifically to represent connections and relationships

By Structure Semi-Structured (or a distinct "Graph Structure“)
By Nature Can contain both Quantitative (e.g., edge weights, node properties) and Qualitative data (e.g., node categories, relationship types).



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Key Relationships

Structured Data can contain both Quantitative and Qualitative data types.

- Example: A customer database table has Numerical columns (Age, Salary - Quantitative) and Categorical columns (Customer Tier - Ordinal, Product Category - Nominal).

Unstructured Data is its own format category but can be analyzed to extract Quantitative and Qualitative insights.

- Example: From customer review Text (Unstructured), we can extract:
  - Sentiment score (Quantitative - Continuous)
  - Product categories mentioned (Qualitative - Nominal)

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Some More Examples of Different Types of Data

- **Tabular Data**
  - Text document of the heights of everyone in this class in inches
  - IRS Data for taxpayers
  - Netflix show data
- **Graph Data**
  - Social networks
  - Course prereqs
  - Highway
- **Geo Data**
  - Flight paths
  - Weather patterns
  - All phones on verizon

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Some More Examples of Different Types of Data

- **Raw Data**
  - Image
  - Video
  - Audio
  - Telemetry
- **Hierarchies (Graphy)**
  - Taxonomy for something
  - Family tree
  - File directory

- **Text**
  - All tweets
  - Chat logs
  - Search history
  - Shakespeare
- **Time Series**
  - People in store
  - Stock prices

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Data Formats

- **CSV / TSV**
- **Image**
  - .jpg
  - .png
- **Audio**
  - .wav
  - .mpg
- **SQL Database**
  - mySQL
  - Postgres
  - etc…

- **No SQL Database**
  - Bigtable
  - Accumulo

- **JSON**
- **XML / HTML**

- We mostly cover Tabular, JSON in this course

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Data Format: TSV / CSV

- **CSV (Comma-Separated Values)** A plain-text format where data values are separated by commas; commonly used for tabular data.

- **TSV (Tab-Separated Values)** A plain-text format where data values are separated by space of tab; commonly used for tabular data.

    - Both are common file formats used to store structured data in a simple tabular format, with rows and columns.
    - Easy to import and export across various data analysis tools and programming languages.
    - Any CSV reader worth anything can parse files with any delimiter, not just a comma (e.g., “TSV” for tab-separated values)

Delimiter: The separator character : the comm (,), the tab (`\t`), colon (`:`) and semi-colon (`;`) characters.

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How data might be structured in CSV Files

```
Artist,Music,Album,Year,Genre,2022,2021,2020,2019,2018,2017,2016,2015

The Black Crowes,Remedy,The Southern Harmony and Musical
Companion,1992,Southern Rock,500,,324,290,132,64,36,

Asia,Only Time Will Tell,Asia,1982,Progressive Rock,499,,,,,,,

Collective Soul,Shine,Hints Allegations and Things Left Unsaid,1993,Alternative
Rock,498,,,419,485,403,,
```

- CSV (Comma-Separated Values) files are a common way to store tabular data.
- Python's pandas library makes it easy to load (df = pd.**read_csv**('data.csv')) and manipulate CSV data.

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CSV File representation in tabular formdata

classic_rock_playlist.csv (39.93 kB)

Detail | Compact | Column

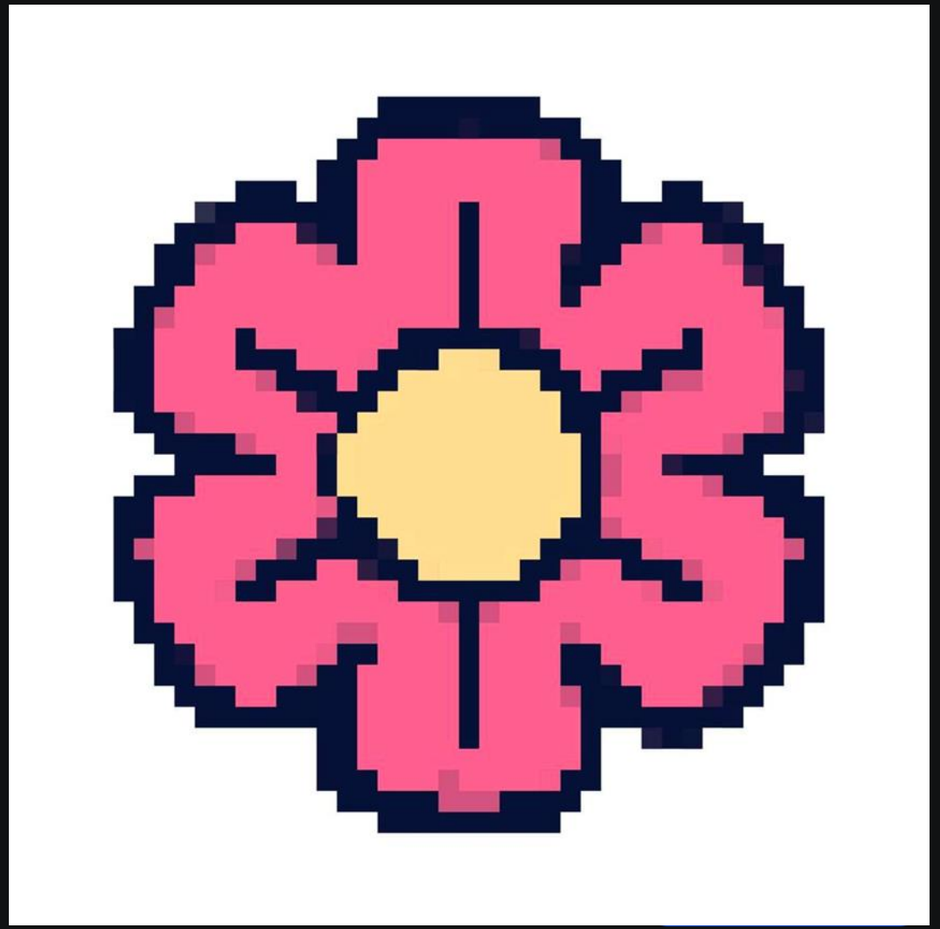
10 of 13 columns

Artist	Music	Album	Year	Genre	2022	2021
The Black Crowes	Remedy	The Southern Harmony and Musical Companion	1992	Southern Rock	500	
Asia	Only Time Will Tell	Asia	1982	Progressive Rock	499	
Collective Soul	Shine	Hints Allegations and Things Left Unsaid	1993	Alternative Rock	498	
Billy Idol	Sweet Sixteen	Whiplash Smile	1986	Rock	497	
Collective Soul	December	Collective Soul	1995	Alternative Rock	496	
Duran Duran	Save a prayer	Rio	1982	Synthpop	495	466
Men at Work	Down Under	Business as Usual	1981	New Wave	494	
Brian Setzer	Summertime Blues	La Bamba soundtrack	1987	Rock and Roll	493	
Simple Minds	Dont You Forget About Me	The Breakfast Club Original Motion Picture Soundtrack	1985	Pop Rock	492	

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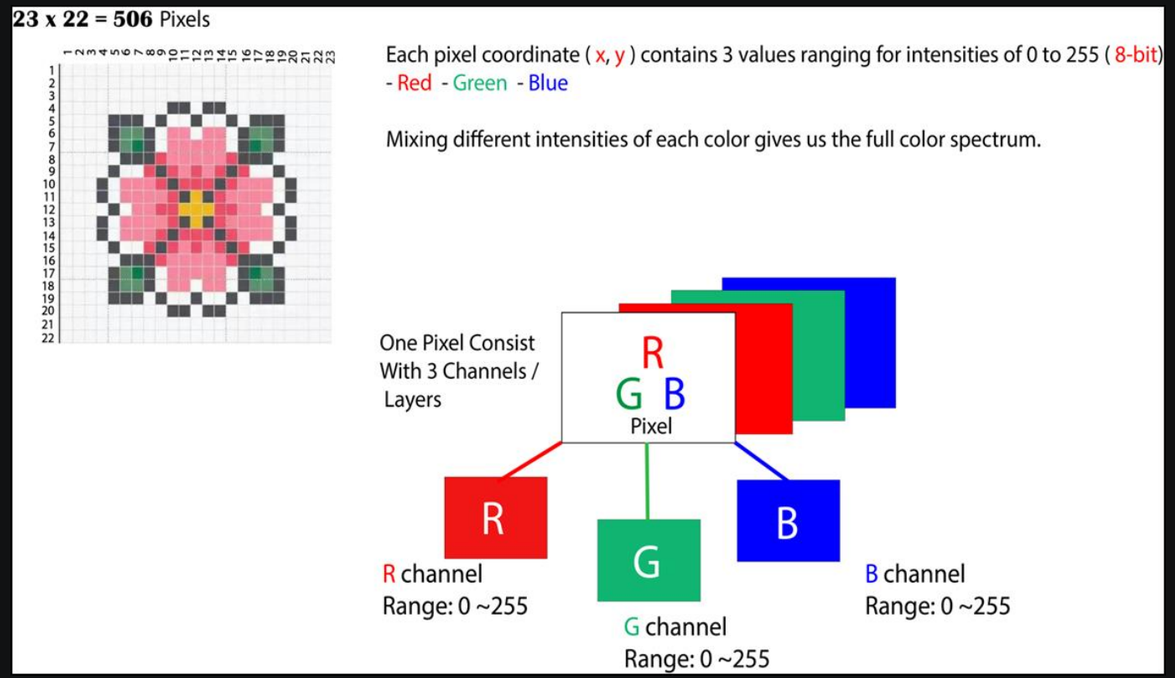
Data Format: Images

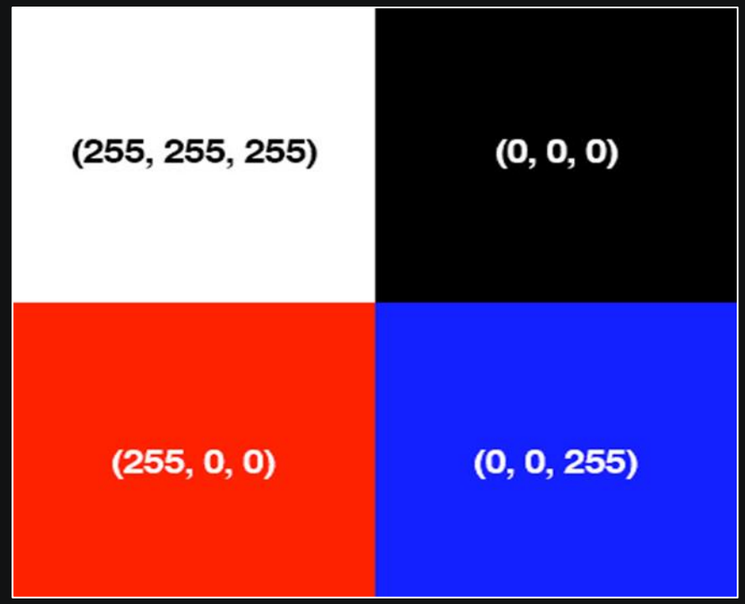
Image data encompasses **the visual content and properties of an image** (visual representation), including details such as **colors, shapes, patterns, and pixel values**.



Data Format: Images

- **Pixel Structure:** Images are organized as grids of pixels (tiny building block).
- **Pixel Data:** Each pixel holds color information.
- **Color Channels:** In color images, pixels have red, green, and blue (RGB) channels.
  - **Channel→ is like a separate layer of information about color.**
- **Channel Values:** Each channel ranges from 0 (no intensity) to 255 (maximum intensity).
- **Intensity Values:** Values between 0 and 255 create varying shades of the color





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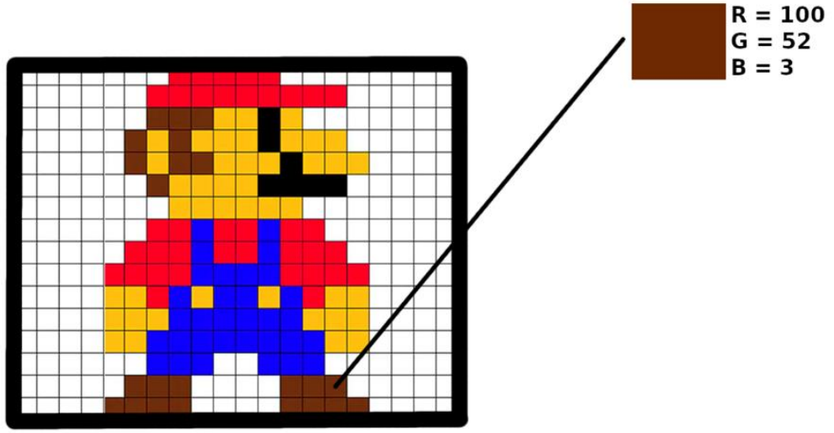
Data Format: Images

Images can be compressed to save storage space.

Two types of compressions:

- **Lossy Compression:** Sacrifices some data to reduce file size.
    - Suitable for photos where minor quality loss is acceptable (e.g., JPEG).
- **Lossless Compression:** Reduces file size without any loss of quality.
    - Ideal for preserving image quality in medical imaging, digital art, or graphics (e.g., PNG, GIF).

→Varying Underlying Resolutions (level of detail and clarity)





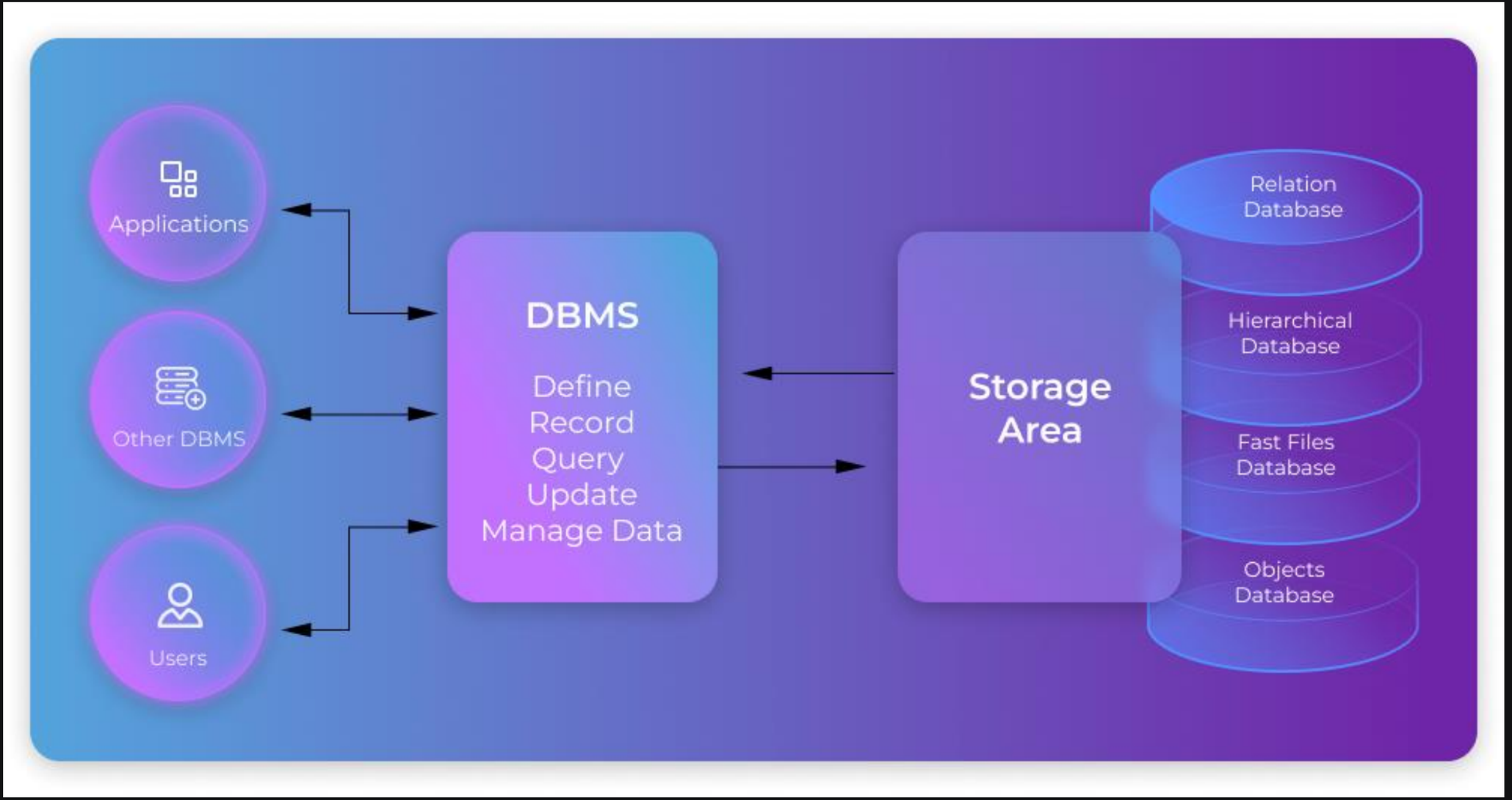
Databases

A database is an organized collection of structured information, or data, typically stored electronically in a computer system.

- Handle more complex data relationships, often organized in tables.
- It **efficiently manages and allows retrieval, updating, and manipulation** of information for various applications.

```
dvdrental=# select title, release_year, length, replacement_cost from film
dvdrental-#   where length > 120 and replacement_cost > 29.50
dvdrental-#   order by title desc;
          title           | release_year | length | replacement_cost
--------------------------+--------------+--------+------------------
 West Lion                |         2006 |    159 |            29.99
 Virgin Daisy             |         2006 |    179 |            29.99
 Uncut Suicides           |         2006 |    172 |            29.99
 Tracy Cider              |         2006 |    142 |            29.99
 Song Hedwig              |         2006 |    165 |            29.99
 Slacker Liaisons         |         2006 |    179 |            29.99
 Sassy Packer             |         2006 |    154 |            29.99
 River Outlaw             |         2006 |    149 |            29.99
 Right Cranes             |         2006 |    153 |            29.99
 Quest Mussolini          |         2006 |    177 |            29.99
 Poseidon Forever         |         2006 |    159 |            29.99
 Loathing Legally         |         2006 |    140 |            29.99
 Lawless Vision           |         2006 |    181 |            29.99
 Jingle Sagebrush         |         2006 |    124 |            29.99
 Jericho Mulan            |         2006 |    171 |            29.99
 Japanese Run             |         2006 |    135 |            29.99
 Gilmore Boiled           |         2006 |    163 |            29.99
 Floats Garden            |         2006 |    145 |            29.99
 Fantasia Park            |         2006 |    131 |            29.99
 Extraordinary Conquerer  |         2006 |    122 |            29.99
 Everyone Craft           |         2006 |    163 |            29.99
 Dirty Ace                |         2006 |    147 |            29.99
 Clyde Theory             |         2006 |    139 |            29.99
 Clockwork Paradise       |         2006 |    143 |            29.99
 Ballroom Mockingbird     |         2006 |    173 |            29.99
(25 rows)
```

Databases



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JSON

- JSON stands for JavaScript Object Notation (a lightweight data interchange format).
- A text format for storing and transporting data; ex: Web APIs
- JSON is "self-describing" and easy to understand
  - It is easy for humans to read and write and easy for machines to parse and generate.
  - popular choice for representing data in web APIs, mobile app development, configuration files, and client-server communication

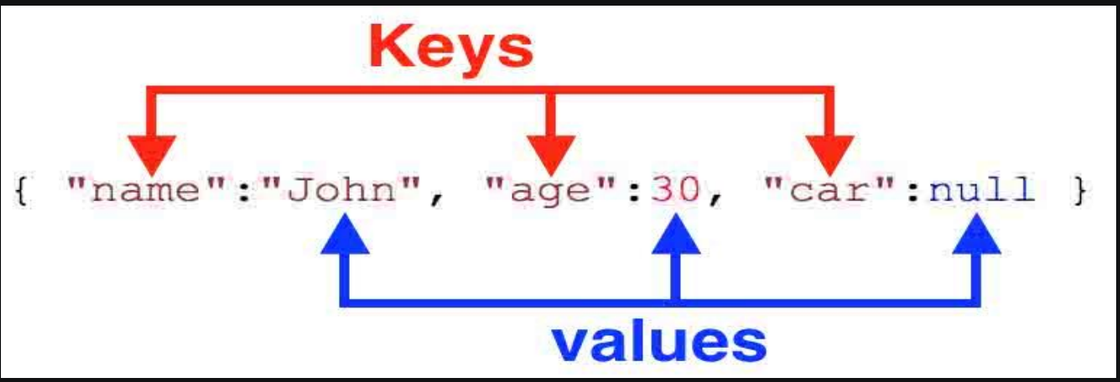
```
'{"name":"John", "age":30, "car":null}'
```

In this Example:

It defines an object with 3 properties:

- name
- age
- car

Each property has a value.



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JSON Structure

- JSON represents data as key-value pairs; organized in a hierarchical structure using objects and arrays
- Objects (collections of key-value pairs),
- Arrays (ordered lists of values)
- JSON accommodates various data types, such as strings, numbers, arrays, objects, booleans, and null, allowing for nested structures.

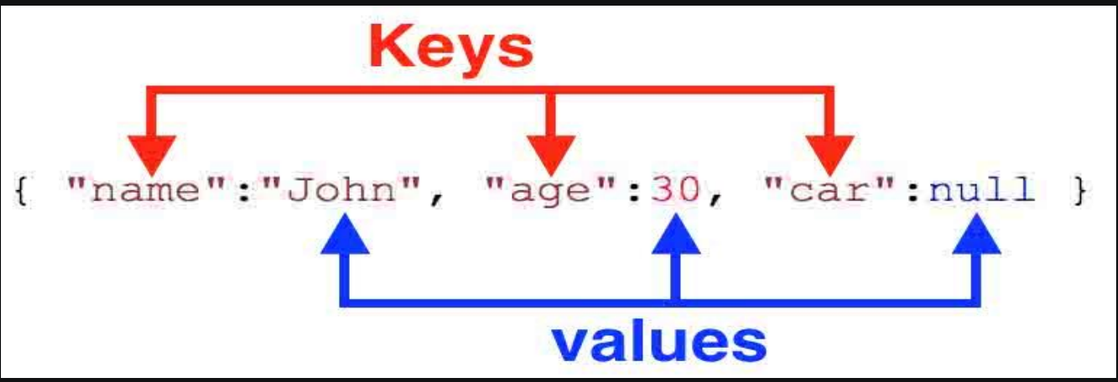
```
'{"name":"John", "age":30, "car":null}'
```

In this Example:

It defines an object with 3 properties:

- name
- age
- car

Each property has a value.



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JSON: Example



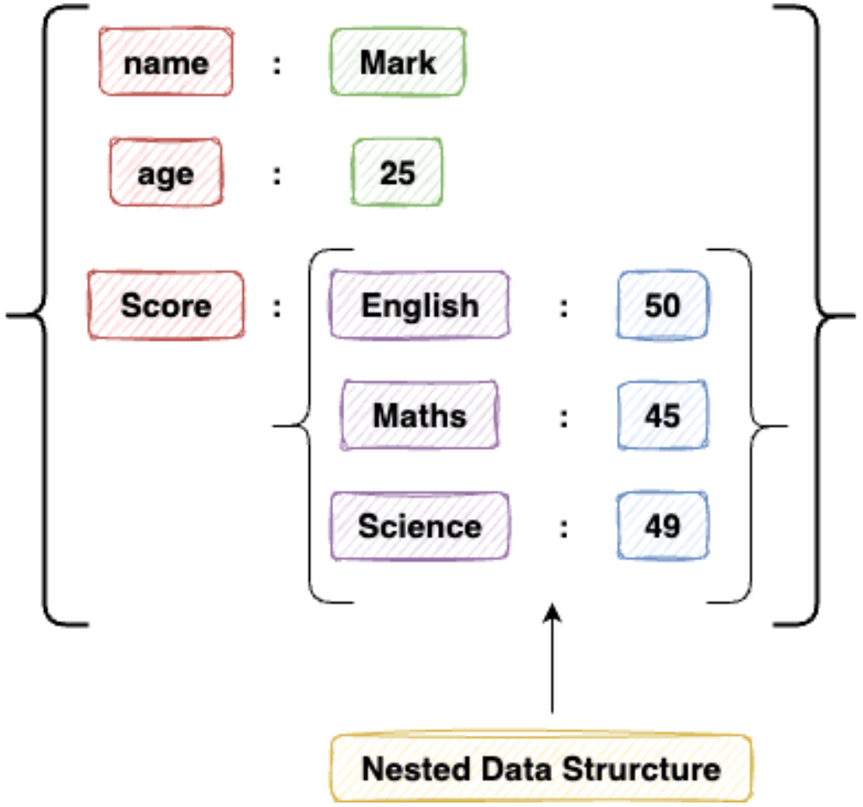
Object

```
{
   creationDate : 2015-11-20T23:19:43.701Z,
   modifiedDate : 2015-11-20T23:19:43.701Z,
   name : demo,
   members : [
      {name : first},
      {name : second},
      {name : third}
   ]
}
```

Name-value pair
comma separated

Array
containing objects with a name-value pair

```
{
  "volume": "blaring",
  "current" : {
              "band": "rednex",
              "song": "cotton eye joe",
              "members":[
                        {"firstname":"Kent","lastname":"Olander"},
                        {"firstname":"Urban","lastname":"Landgren"},
                        {"firstname":"Jonas","lastname":"Lundstrom"},
                        {"firstname":"Tor","lastname":"Nilsson"}
              ]
             },
  "next" : {
          "band": "the dubliners",
          "song": "finnegan's wake",
          "members":[
                    {"firstname":"Ronnie","lastname":"Drew"},
                    {"firstname":"Luke","lastname":"Kelly"},
                    {"firstname":"Ciaran","lastname":"Bourke"},
                    {"firstname":"Barney","lastname":"McKenna"}
          ]
       }
}
```



JSON in Python

In Python, the json module is commonly used to work with JSON data.

- Use **json.dumps()** to convert Python objects to JSON format
- Use **json.loads()** to convert JSON data back to Python objects.

```
# Python object
data = {
    'name': 'John Doe',
    'age': 30,
    'city': 'New York',
    'skills': ['JavaScript', 'Python', 'SQL']
}

# Convert Python object to JSON
json_data = json.dumps(data, indent=2)
print("JSON Data:")
print(json_data)

# Convert JSON data to Python object
parsed_data = json.loads(json_data)
print("\nParsed Data:")
print(parsed_data)
```

XML / HTML

- HTML →Hypertext Markup Language
- XML → eXtensible Markup Language
- Both are markup languages (define the text document within tag which defines the structure of web pages).
- Used for structuring and organizing content on the web and other digital documents, but they serve different purposes:

HTML

```
<!DOCTYPE html>
<html>
    <head>
        <title>Example</title>
        <link rel="stylesheet" href="st
    </head>
    <body>
        <h1>
            <a href="/">Header</a>
        </h1>
        <nav>
            <a href="one/">One</a>
            <a href="two/">Two</a>
            <a href="three/">Three</a>
        </nav>
```

```
<person>
    <address>
        <first_name>Peter</first_name>
        <last_name>Miller</last_name>
        <street>Hauptstrasse</street>
        <number>20</number>
        <city>Zurich</city>
    </address>
</person>
```

HTML

- HTML is primarily used for creating web pages and presenting content on the internet.
- HTML has a fixed set of predefined tags for structuring content, emphasizing elements like headings, paragraphs, links, etc.



```
<!DOCTYPE html>
<html>
    <head>
        <title>Example</title>
        <link rel="stylesheet" href="st
    </head>
    <body>
        <h1>
            <a href="/">Header</a>
        </h1>
        <nav>
            <a href="one/">One</a>
            <a href="two/">Two</a>
            <a href="three/">Three</a>
        </nav>
```

XML

- XML is designed to transport and store data, with a focus on describing the structure of the data and is not concerned with presentation.
- XML allows users to define custom tags, allowing flexibility to create custom structures

In this Example:

```
<person>
    <address>
        <first_name>Peter</first_name>
        <last_name>Miller</last_name>
        <street>Hauptstrasse</street>
        <number>20</number>
        <city>Zurich</city>
    </address>
</person>
```

How to Get Data?

- Given to you by your company
- Gathered from databases
- From the internet
- Using a restful API

In this Example:

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Beautiful Soup and Parsing HTML

Beautiful Soup is a Python library for parsing HTML and XML, making web scraping (extracting data directly from web pages) and data extraction easier.

```
soup = BeautifulSoup(page.content, 'html.parser')
soup.find_all('p')
```

```
[<p>Here is some simple content for this page.</p>]
```

Note that `find_all` returns a list, so we'll have to loop through, or use list indexing, it to extract text:

```
soup.find_all('p')[0].get_text()
```

```
'Here is some simple content for this page.'
```

Notes: Don't write own parser. Install Beautiful soup and Use Beautiful Soup to parse HTML content by creating a Beautiful Soup object.

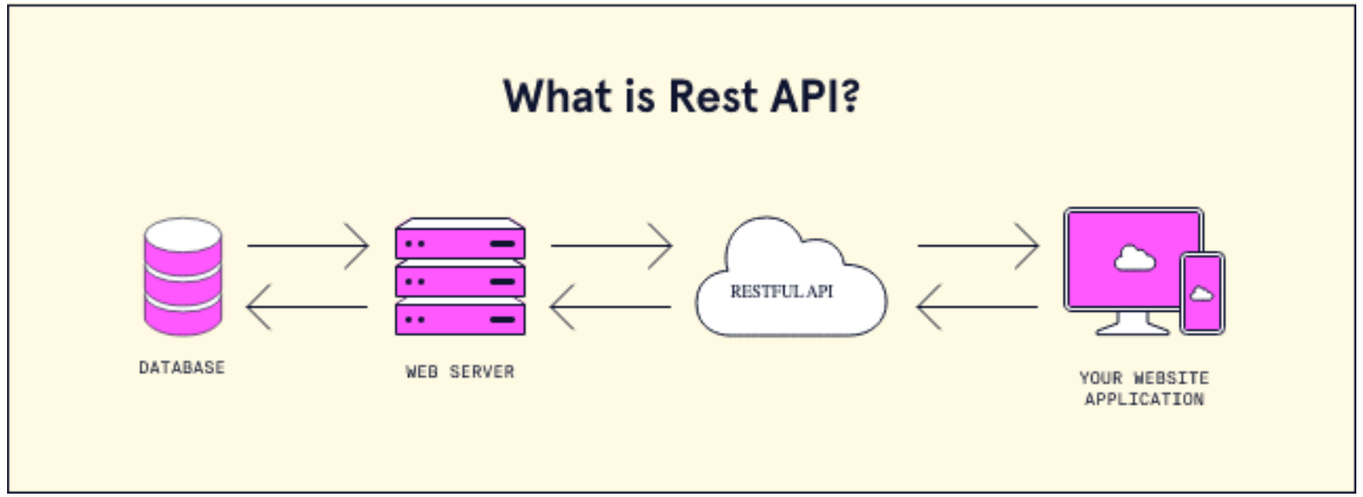
Restful APIs (Application Programming Interface)

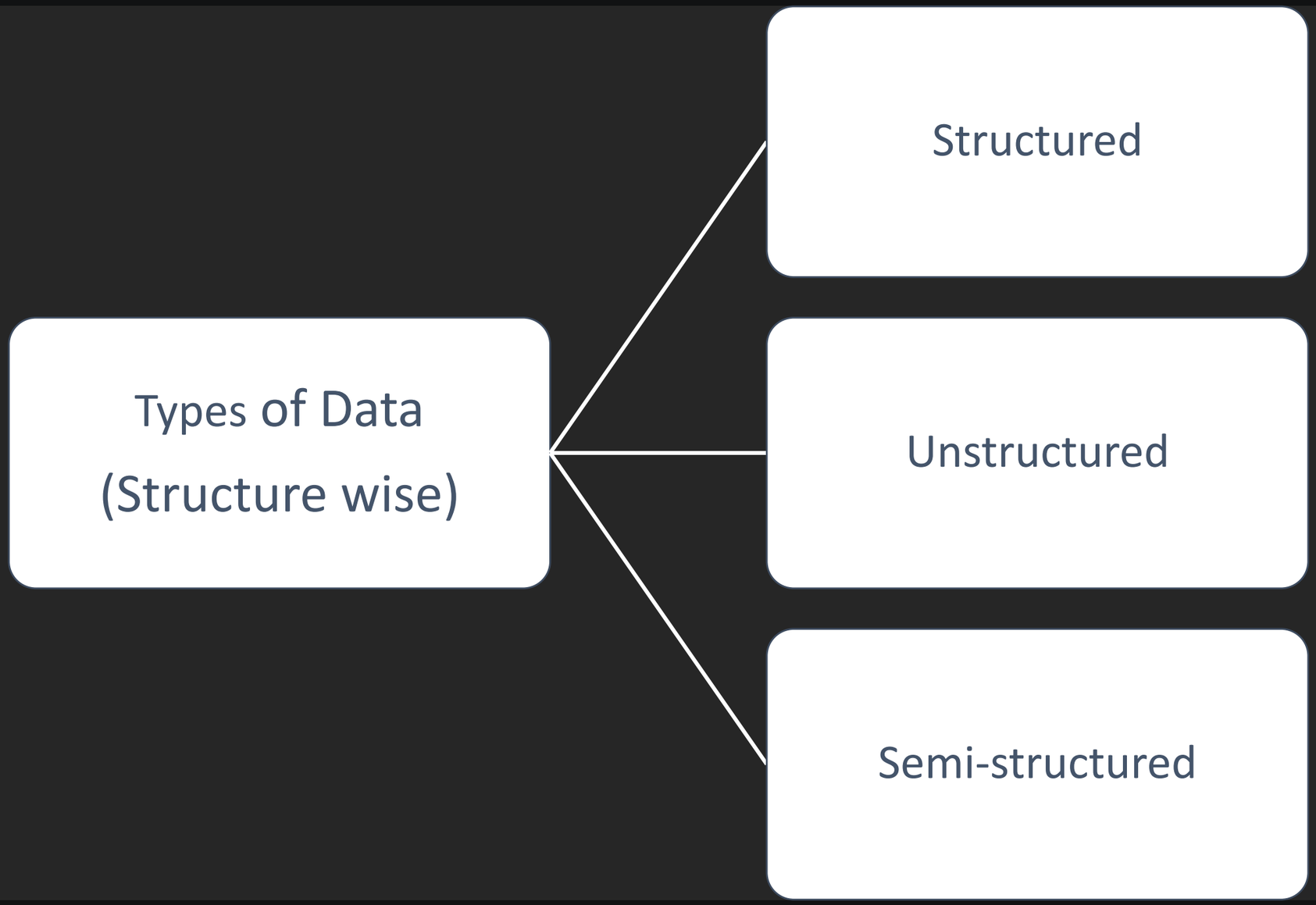
RESTful API (Representational State Transfer API) provide **a structured and documented way** to access data from websites or services that offer them.

- APIs are a reliable way to access web data.
- APIs create a service agreement between applications.
- They enable communication via requests and responses.
- API documentation guides effective usage and data interpretation.effectively, interpret the received data correctly etc.

```
import requests
response = requests.get("http://api.open-notify.org/astros.json")
print(response)
```

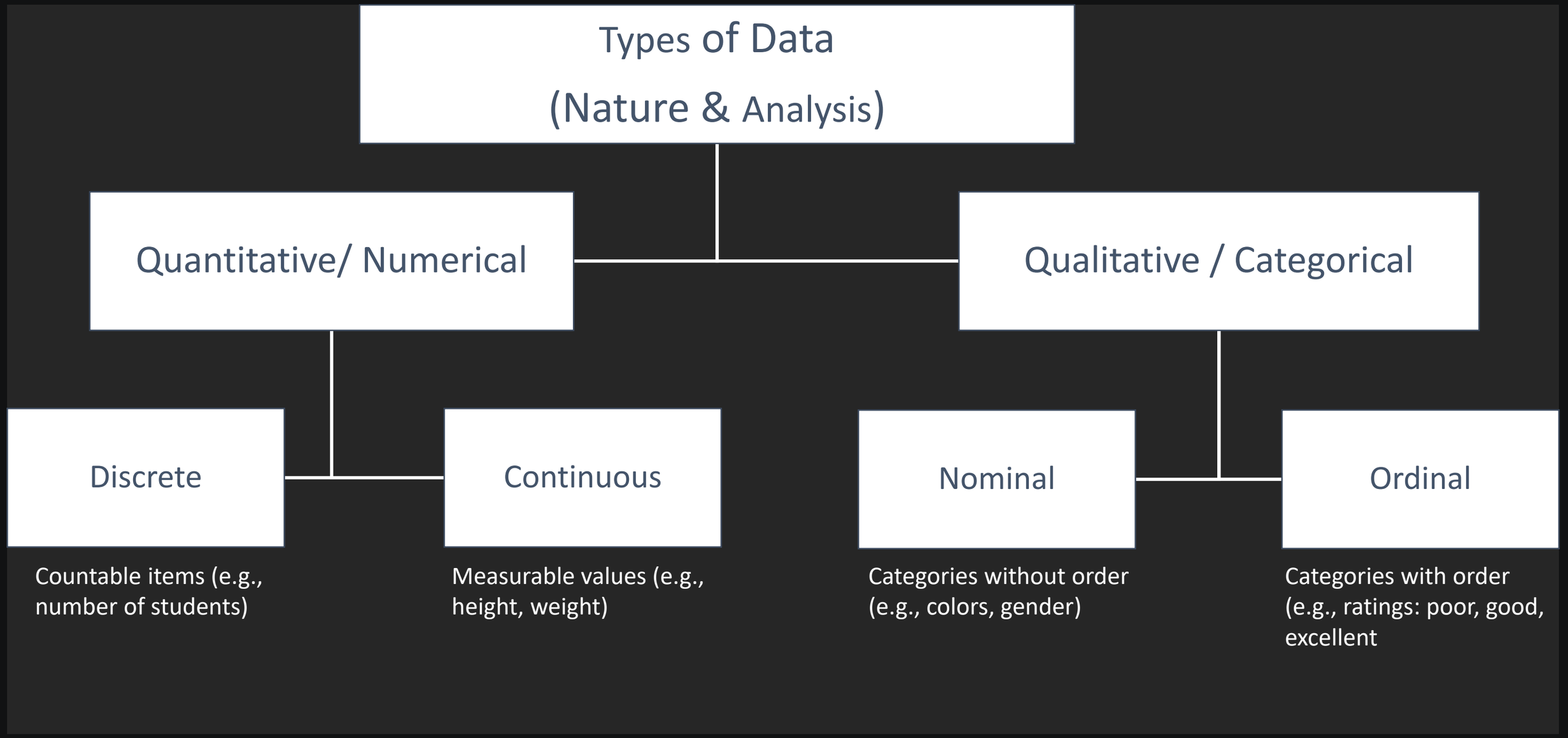
"If you send me a specific request, I will return some information in a structured and documented format."





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Types of Data (Nature & Analysis)



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Discrete vs. Continuous Attributes

- **Discrete** Attribute
  - Has only a **finite** or **countable infinite set of values**
  - Examples: zip codes, counts, or the set of words in a collection of documents
  - Often represented as **integer variables.**
  - Note: binary attributes are a special case of discrete attributes

- **Continuous** Attribute
  - Has **real numbers** as attribute values
  - Examples: temperature, height, or weight.
  - Practically, real values can only be measured and represented using a **finite number of digits.**
  - Continuous attributes are typically represented as **floating-point variables.**

Types of Attribute Values

- Nominal
  - E.g., profession, ID numbers, eye color, zip codes
- Ordinal
  - E.g., rankings (e.g., army, professions), grades, height in {tall, medium, short}
- Binary
  - E.g., medical test (positive vs. negative)
- Interval
  - E.g., calendar dates, body temperatures
- Ratio
  - E.g., temperature in Kelvin, length, time, counts

Nominal

- **Categorical / Qualitative**
- Nominal values provide only **enough information** to **distinguish** one object from another
- Examples are eye colour, NRIC number, Postal codes, marital status
- The attribute value has **distinctiveness**. It means that you can check whether an attribute is **equal or not equal to a value**.
- Operators are: =, < >
- Any transformation done must have a **one to one mapping** and result with a permutation of values

```
Original: Small, Medium, Large
Transformed: 1, 2, 3
```

Binary

- **Categorical/Qualitative**
- Nominal attribute with only **2 states** (**0 and 1**)
- 2 types of binary attribute
  - **Symmetric** binary: both outcomes **equally important**
    - E.g., gender
  - **Asymmetric** binary: outcomes **not equally important.**
    - E.g., medical test (positive vs. negative)
    - Convention: assign 1 to most important outcome (e.g., HIV positive)

Ordinal

- **Categorical/Qualitative**
- Ordinal values provide only **enough information** to **distinguish** one object from another and **order** the objects
- E.g., length = {short, medium, long}, exam grades
- The attribute value has **distinctiveness** and **order** . It means that you can check whether an attribute is **equal to, not equal to, greater than or smaller than a value.**
- Operators are: **=, <>, > , <**
- Any transformation done must **ensure the order is preserved**
- Transformation formula : **new_value = f(old_value)**. Must be a **monotonic** function.
  - {short, medium, long} ⟶ {1,2,3}

Interval

- **Numeric/Quantitative**
- Measured on a **scale** **of equal-sized units**
- **Differences between attribute values are meaningful**
- E.g., calendar dates, temperature in Fahrenheit /Celsius
- The attribute value can be **added or subtracted** from a value
- Operators are: **=, <>, > , < , +, -**
- Transformation formula :**new_value = a * old_value + b**
  - F = (9/5) C + 32; F = Fahrenheit, C = Celsius

Ratio

- **Numeric/Quantitative**
- Both **differences and ratio between attribute values are meaningful**
- Values as being **an order of magnitude larger than the unit of measurement**
- E.g., age, temperature in Kelvin, mass, length
- The attribute value can be **added, subtracted or multiplied** from a value
- Operators are: **=, <>, > , < , +, -, * , /**
- Transformation formula : **new_value = a * old_value**
  - 1 hour = 3600 seconds
  - 2 hours is always twice of 1 hour
  - 10 K° is twice as high as 5 K°

Summary

- **Types of data:** Various data types (tabular, text, graph, unstructured) impact data preparation in the data science lifecycle.
- **File Formats:** Knowledge of file formats (e.g., CSV, JSON etc.) aids data ingestion and transformation during data preparation.
- **Databases:** are structured repositories for storing and retrieving data, playing a central role in data management during the data science process.
- **Data Acquisition:** RESTful APIs and web scraping are some key for gathering data at the beginning of the data science lifecycle.

Understanding these elements is crucial for progressing through the data science lifecycle, leading to data-driven insights and solutions.

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Thank You

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